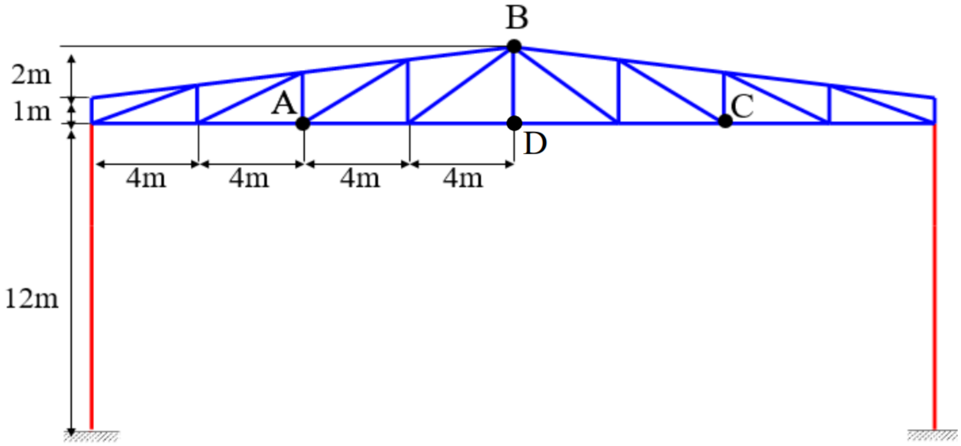
Dynamics of Mechanical Systems Academic year 2022-2023

Yearwork



Consider the structure shown above, representing a hangar portal frame. All beams are made of steel ($E=2.06e11$ N/m^2, $\rho=7800$ kg/m^3). Beams in blue colour have IPE240 cross section ($A=3.912e-3$ m^2, $I_y=3.892e-5$ m^4) and beams in red colour have IPE500 cross section ($A=0.01155$ m^2, $I_y=4.82e-4$ m^4). Damping is defined according to the proportional damping assumption: $[C]=\alpha[M]+\beta[K]$, with $\alpha=0.2$ s^{-1} and $\beta=8.0e-5$ s.

POINT1

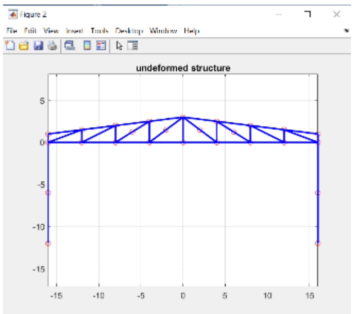
Firstly we needed to define the max length of each profile of beam used, in particular we have 2 different profiles so we have to compute two max lengths according to the condition:

$$\bar{\Omega}_{1min,j} = c 2\pi \bar{f}_{max} \quad \rightarrow \quad L_{max,j} = \sqrt{\frac{\pi^2}{\bar{\Omega}_{1min,j}} \sqrt{\frac{EI_j}{m_j}}}$$

obtaining L240max = 4.47 m and L500max = 6.4 m.

So now we can define FE and nodal points, considering the beams which are longer than L max. Now It is possible to define the .inp file in matlab considering the location of node, the beam properties and damping properties.

```
%% descrizione file input .inp

%IPE 240 roof + IPE 500 pillars

*NODES
1       0 0 0                   -16   0
2       0 0 0                   -12   0
3       0 0 0                    -8   0
4       0 0 0                    -4   0
5       0 0 0                     0   0
6       0 0 0                     4   0
7       0 0 0                     8   0
8       0 0 0                    12   0

9       0 0 0                    16   0

10      0 0 0          16   1
11      0 0 0           12   1.5
12      0 0 0           8   2
13      0 0 0           6 1.25
14      0 0 0           4   2.5
15      0 0 0           2 1.5
16      0 0 0           0   3
17      0 0 0       -2   1.5
18      0 0 0          -4   2.5
19      0 0 0          -6 1.25
20      0 0 0          -8   2
21      0 0 0          -12   1.5
22      0 0 0          -16  1
23      0 0 0          -16 -6
24      0 0 0           16 -6
25      1 1 1          -16 -12
26      1 1 1           16  -12
*ENDNODES

! beam nr.   i-th node nr.  j-th node nr.
mass [kg/m]   EA [N]  EJ [Nm^2]
! EAb = 2.3793e+09
! EAb = 8.0587e+08

*BEAMS
1   1   2  30.7 8.0587e+08 8017520
2   2   3  30.7 8.0587e+08 8017520
3   3   4  30.7 8.0587e+08 8017520
4   4   5  30.7 8.0587e+08 8017520
5   5   6  30.7 8.0587e+08 8017520
6   6   7  30.7 8.0587e+08 8017520
7   7   8  30.7 8.0587e+08 8017520
8   8   9  30.7 8.0587e+08 8017520
9   9   10 30.7 8.0587e+08 8017520
10  10  11 30.7 8.0587e+08 8017520
11  9   11 30.7 8.0587e+08 8017520
12  11  12 30.7 8.0587e+08 8017520
13  8   12 30.7 8.0587e+08 8017520
14  12  14 30.7 8.0587e+08 8017520
15  7   13 30.7 8.0587e+08 8017520
16  13  14 30.7 8.0587e+08 8017520
17  14  16 30.7 8.0587e+08 8017520
18  6   15 30.7 8.0587e+08 8017520
19  15  16 30.7 8.0587e+08 8017520
20  5   16 30.7 8.0587e+08 8017520
21  16  17 30.7 8.0587e+08 8017520
22  4   17 30.7 8.0587e+08 8017520
23  16  18 30.7 8.0587e+08 8017520
24  18  19 30.7 8.0587e+08 8017520
25   3  19 30.7 8.0587e+08 8017520
26  18  20 30.7 8.0587e+08 8017520
27  2   20 30.7 8.0587e+08 8017520
28  20  21 30.7 8.0587e+08 8017520
29  21  1  30.7 8.0587e+08 8017520
30  21  22 30.7 8.0587e+08 8017520
31  1   22 30.7 8.0587e+08 8017520
32  2   21 30.7 8.0587e+08 8017520
33  3   20 30.7 8.0587e+08 8017520
34  4   18 30.7 8.0587e+08 8017520
35  6   14 30.7 8.0587e+08 8017520
36  7   12 30.7 8.0587e+08 8017520
37  8   11 30.7 8.0587e+08 8017520
38  1   23 90.7 2.3793e+09 99292000
39  23  25 90.7 2.3793e+09 99292000
40  9   24 90.7 2.3793e+09 99292000
41  24  26 90.7 2.3793e+09 99292000
*ENDBEAMS

*DAMPING
0.2 8.0e-5
```

Undeformed structure: 

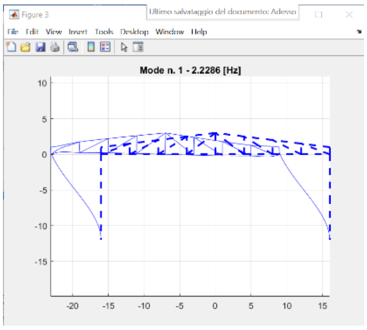
POINT2/3

Once the input file, which defines the structure, is ready to be used we can run dmb_fem2 software to perform the frequency analysis in particular:

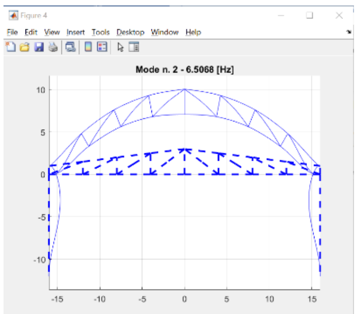
2. Compute and print the first 4 natural frequencies of the structure and their modal shapes.
3. Compute and print the Bode diagrams of the frequency response functions for a unit input force applied at point A in vertical direction considering as output the horizontal and vertical displacements of point B. Compute and print the Bode diagrams (in linear scale) of the frequency response functions for the same input force considering as outputs the horizontal and vertical acceleration of point B.

First 4 modes:

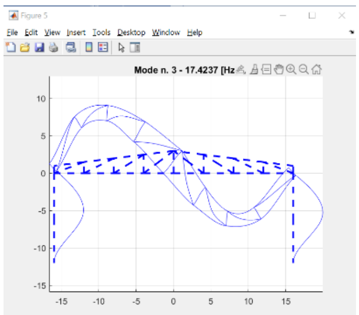
- Mode1: nat_freq_1 = 2.23 Hz



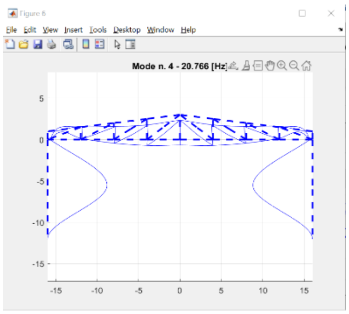
- Mode2: nat_freq_2 = 6.5 Hz



- Mode3: nat_freq_3 = 17.42 Hz



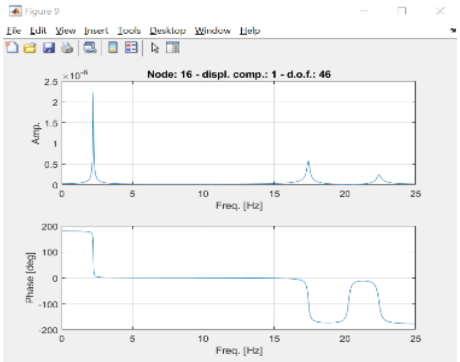
- Mode4: nat_freq_4 = 20 Hz



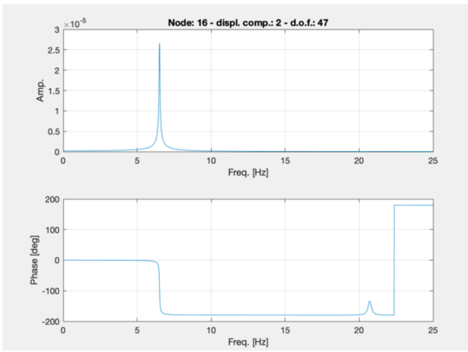
Bode diagram of point 3

Using dmb_fem2 we can also compute the FRF of an input force applied on the system on one of its nodes. (graphics are computed up to 25Hz)

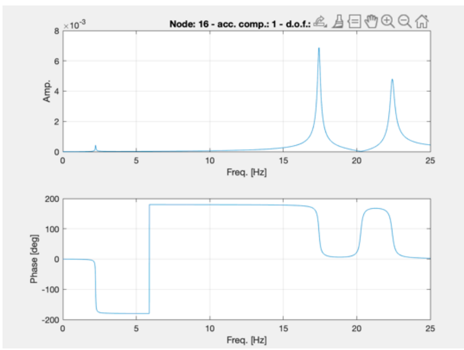
- Horizontal displacement of the point B given input vertical force in point A:



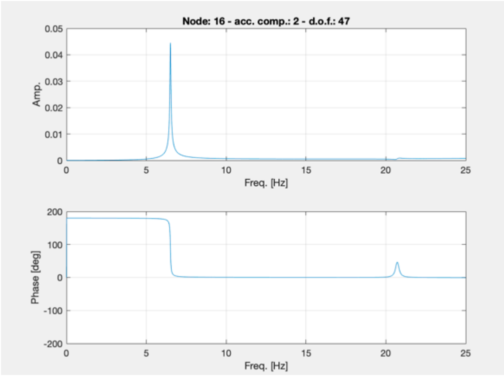
- Vertical displacement of the point B given input vertical force in point A:



- Horizontal acceleration of point B given input vertical force in point A:



- Vertical acceleration of point B given input vertical force in point A:



POINT4

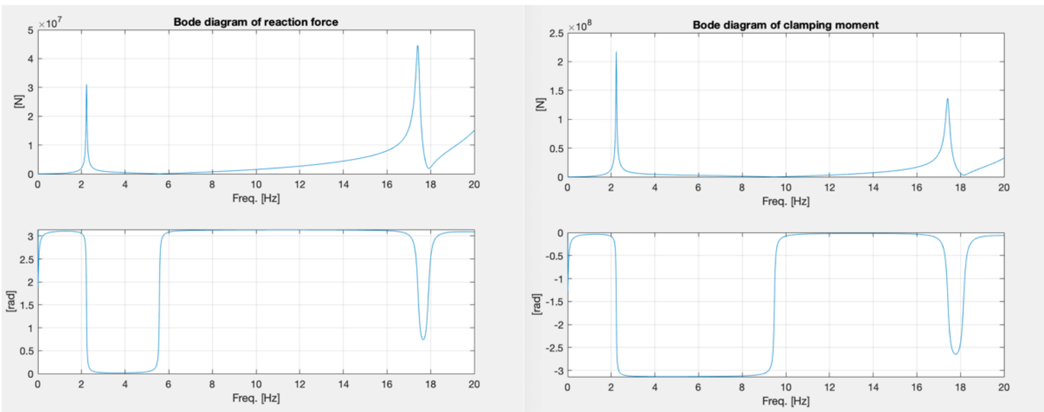
Now we have to perform some detailed analysis extracting matrices from dmb_fem2 to matlab. In order to do so we extract a file .mat containing all the matrices (mass, damping and stiffness) and we load it in the matlab script in order to perform matrix partition the FRF required by point 4.

- ***Matrices partition:***

```
%% MATRICES PARTITION

load beam_FEM_v1_mkr.mat;

% the last six rows-colomns of each matrix are related to the constrained
% nodes 25 and 26, so:

MFF= M(1:72,  1:72);
CFF= R(1:72,  1:72);
KFF= K(1:72,  1:72);

MFC= M(1:72,  73:78);
CFC= R(1:72,  73:78);
KFC= K(1:72,  73:78);

MCF= M(73:78,  1:72);
CCF= R(73:78,  1:72);
KCF= K(73:78,  1:72);

MCC= M(73:78, 73:78);
CCC= R(73:78, 73:78);
KCC= K(73:78, 73:78);

% natural frequencies and modes of vibration

[eigenvectors eigenvalues]=eig(MFF\KFF);

freq=sqrt(diag(eigenvalues))/2/pi
eigenvectors
```

- Compute and print the ***Bode diagrams of the frequency response functions*** for a unit horizontal displacement applied in phase at the base of both pillars, considering as output the horizontal reaction force and clamping moment in the clamp of the left pillar:

```
%% point 4: FRF1_ constraint displacement

i= sqrt(-1);
vett_f= 0:0.1:20;
F0=0;
y0= [1 0 0 1 0 0]';

for k=1:length(vett_f)
    ome=2*pi*vett_f(k);
    A=-ome^2*MFF+i*ome*CFF+KFF;
    Q0FC=-(-ome^2*MFC+i*ome*CFC+KFC)*y0;
    x0=A\Q0FC;
    %x0dd=-ome^2*x0;
    %x0d=ome*i*x0;
    Reaction_f= (-ome^2*MCC+i*ome*CCC+KCC)*y0+(-ome^2*MCF+i*ome*CCF+KCF)*x0;

    out1=Reaction_f(1); %horizontal reaction force in the left pillar
    out2= Reaction_f(3); %clamping moment in the left pillar

    mod1(k)=abs(out1);
    phase1(k)=angle(out1);
    mod2(k)=abs(out2);
    phase2(k)=angle(out2);

end

figure
subplot 211;plot(vett_f,mod1);grid;xlabel('Freq. [Hz]');ylabel('[N]');title('Bode diagram of reaction force')
subplot 212;plot(vett_f,phase1);grid;xlabel('Freq. [Hz]');ylabel('[rad]')

figure
subplot 211;plot(vett_f,mod2);grid;xlabel('Freq. [Hz]');ylabel('[N]');title('Bode diagram of clamping moment')
subplot 212;plot(vett_f,phase2);grid;xlabel('Freq. [Hz]');ylabel('[rad]')
```



(Remember the values of the clamping moment, because it is useful for point 7 of the yearwork.)

POINT5

1. Build the waveform we want to use as input.

   Considering the periodic function obtained by the sum of cos and sin functions, being the periodic function odd, all the Aks are going to be 0.
   In order to obtain the values of the Bks, knowing that the freq range of interest is the one from 0 to 20Hz, we'll consider only the first 6 elements (being the distance between them of 3.3Hz).

1 Periodic functions: Fourier Series

Let *f(t)* be a periodic function of time with period T:

$$f(t) = f(t+T)$$

The function can be expanded in a Fourier series:

$$f(t) = \tfrac{1}{2}a_0 + \sum_{k=1}^{\infty}\left[a_k \cos(k\Omega_0 t) + b_k \sin(k\Omega_0 t)\right] \tag{1}$$

with:

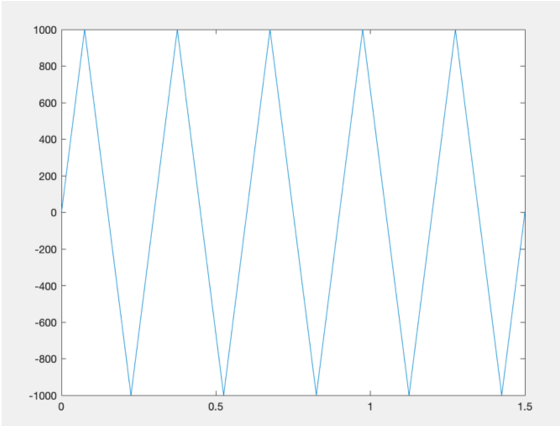
$$\Omega_0 = \frac{2\pi}{T} \tag{2}$$

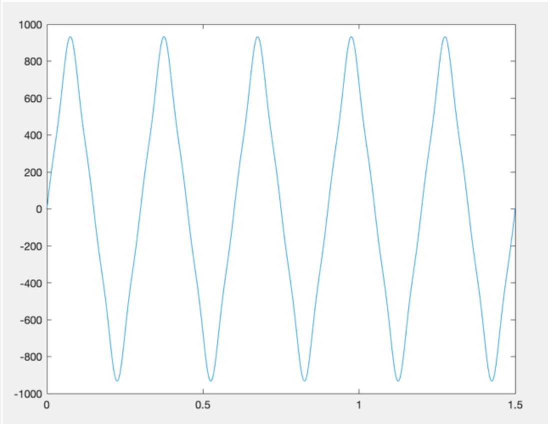
and:

$$\begin{aligned} a_k &= \tfrac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\cos(k\Omega_0 t)\,dt \qquad k = 0,1,2,\ldots \\ b_k &= \tfrac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\sin(k\Omega_0 t)\,dt \qquad k = 1,2,\ldots \end{aligned} \tag{3}$$

Fourier transform of the periodic function:

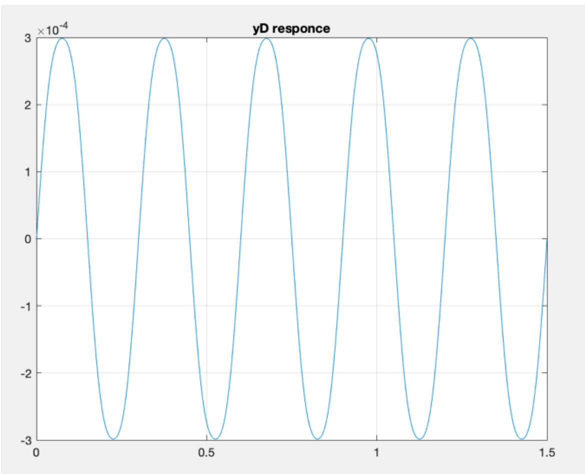
```
T=0.3;
omega=2*pi/T;
t=0:0.001:1.5;
ydd=0;
i=sqrt(-1);
s=1/2;
W=1000*sawtooth(2*pi/0.3*(t+0.075),s); %triangular function
figure; plot(t,W);

A = 1000/(T/4);
f_input = 0;
for k = 1:6
    t1 =  0:0.00001:T/4;
    t2 = T/4:0.00001:T/2;
    f1 = A*t1.*sin(k*omega*t1);
    f2 = A*(T/4)*(1-(4/T*(t2-T/4))).*sin(k*omega*t2);  %
    b(k) = 4/T*(trapz(t1,f1)+trapz(t2,f2));

end

for k=1:6
    f_input= b(k)*sin(k*omega*t) + f_input;
end

figure
plot(t,f_input);title('input Force'); %fourier transform of the triangular function
```

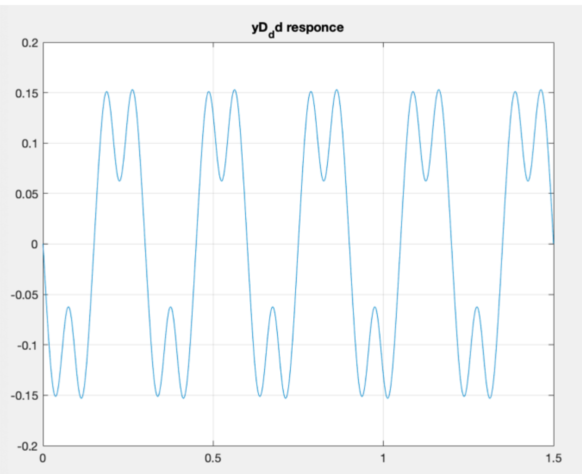




2. Time history of displacement and acceleration of point D:

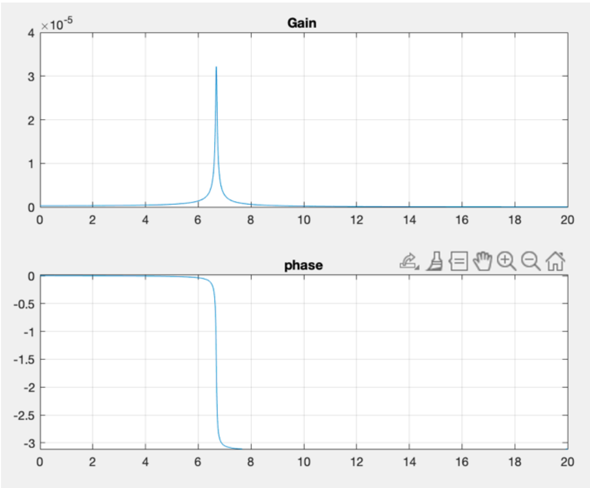
```
F0=zeros(72,1);
F0(14)=f_input(1);
xf0=KFF\F0;          % from the theory X_0 = F0/K = 0 in this case;
y = 0;
ydd=0;
Q0 = zeros(72,1);

for k=1:6
    Q0(14) = b(k)*exp(-i*pi/2);
    omegaj= k*omega;
    A=-omegaj^2*MFF+i*omegaj*CFF+KFF;
    xf0= A\Q0;
    y=y+abs(xf0(14))*cos(omegaj*t+angle(xf0(14)))
    xdd= -omegaj^2*xf0;    %second derivative
    ydd=ydd+abs(xdd(14))*cos(omegaj*t+angle(xdd(14))); %time domain

end

figure; plot(t,y);grid;  title('yD responce');
figure; plot(t,ydd);grid; title('yD_dd responce');
```





3. ***Resonance***

```
vett_f = 0:0.01:20;
Q0(14) = 1;
for k = 1:length(vett_f)
    ome= vett_f(k)*2*pi;
    A = -ome^2*MFF+i*ome*CFF+KFF;
    vect_x_F1=A\Q0;   %vect_x_F0 represents the motion of 24 nodes = 72DOF
    out1 = vect_x_F1(14);

    mod1(k) = abs(out1);
    mod2(k) = angle(out1);
end

figure;
subplot 211 ; plot(vett_f, mod1);grid; title('Gain ');
subplot 212 ; plot(vett_f, mod2);grid; title('phase');
```



Appliying a unitary force on the 5th node (point D) on vertical direction, we can see that the resonance condition will be in the range of frequencies [6,5Hz;7Hz], it means that the values of the input's period for which I'll have resonance are 0,143s<T<= 0,154s.

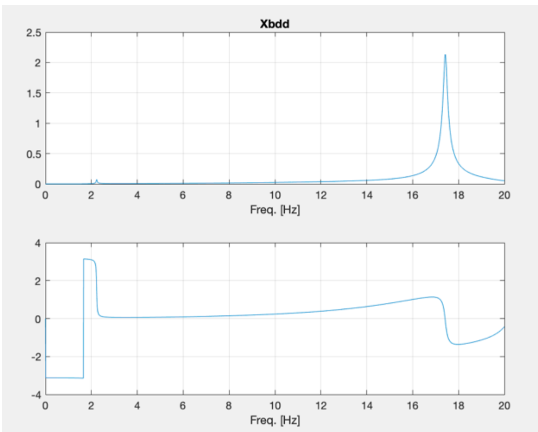
POINT6

6. Plot the Bode diagrams of the horizontal and vertical acceleration of points B and C for the excitation produced by an unbalanced mass **m_s=5kg** having eccentricity ε=5mm and rotating around point A with angular speed $\Omega=2\pi f$ (f=[0:0.01:20]). You can get a hint on how to solve this point from Section 3.3 in:

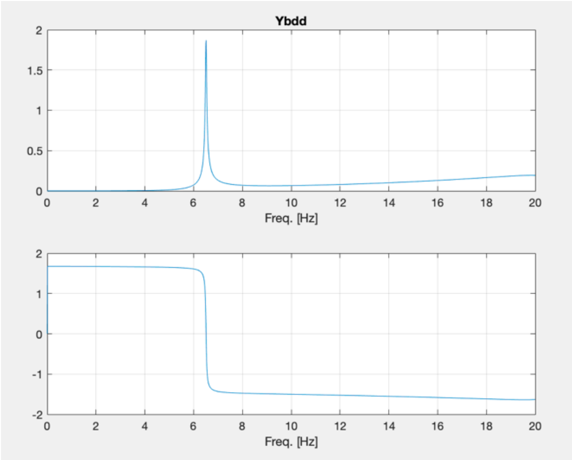
In this case we decide to consider the presence of this mass with eccentricity **e** rotating around point A as two different harmonic excitation [sin, cos] which can be representative of the centrifugal forces we introduce with the mass ms. So the FRF can be computed in this way:

```
i = sqrt(-1);
f = 0:0.01:20;
ec = 0.005;
ms = 5;
for j = 1:length(f)
        ome = 2*pi*f(j);
        A = -ome^2*MFF+i*ome*CFF+KFF;
        Ff = zeros(72,1);
        Ff(7)=ec*ome^2*ms;
        Ff(8)=ec*ome^2*ms*(exp(-i*pi/2));
        x0 = A\Ff;

        % I want nodes 16 and 7 so dof form idb number 19, 20, 46, 47
        xb = x0(46);
        yb = x0(47);
        xc = x0(19);
        yc = x0(20);

        xb_dd = -ome^2*xb;
        yb_dd = -ome^2*yb;
        xc_dd = -ome^2*xc;
        yc_dd = -ome^2*yc;

        mod3(j) = abs(xb_dd);
        mod4(j) = abs(yb_dd);
        fas3(j) = angle(xb_dd);
        fas4(j) = angle(yb_dd);
        mod5(j) = abs(xc_dd);
        mod6(j) = abs(yc_dd);
        fas5(j) = angle(xc_dd);
        fas6(j) = angle(yc_dd);
end
figure()
subplot 211;plot(f,mod3);grid;title('Xbdd');xlabel('Freq. [Hz]')
subplot 212;plot(f,fas3);grid;xlabel('Freq. [Hz]');

figure()
subplot 211;plot(f,mod4);grid;title('Ybdd');xlabel('Freq. [Hz]');
subplot 212;plot(f,fas4);grid;xlabel('Freq. [Hz]');

figure()
subplot 211;plot(f,mod5);title('Xcdd');xlabel('Freq. [Hz]');grid;
subplot 212;plot(f,fas5);grid;xlabel('Freq. [Hz]');

figure()
subplot 211;plot(f,mod6);title('Ycdd');grid;xlabel('Freq. [Hz]');
subplot 212;plot(f,fas6);grid;xlabel('Freq. [Hz]');
```

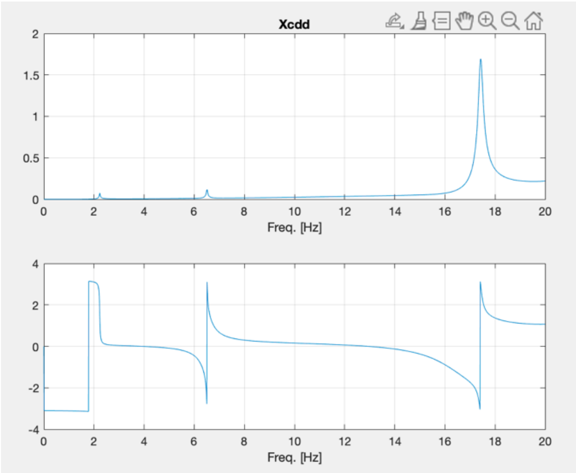
- ***Horizontal acceleration of point B:***



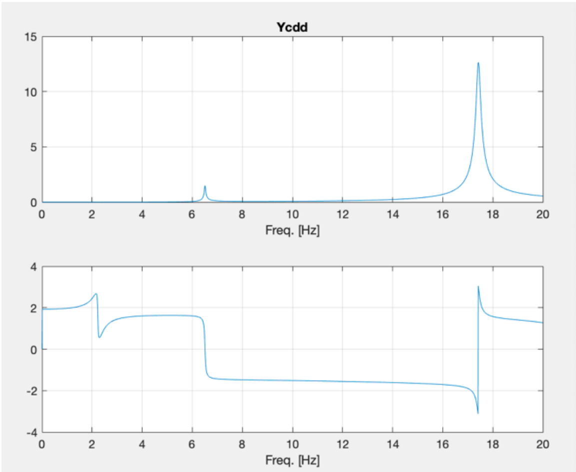
- ***Vertical acceleration of point B:***



- ***Horizontal acceleration of point C:***



- ***Vertical acceleration of point C:***



POINT7

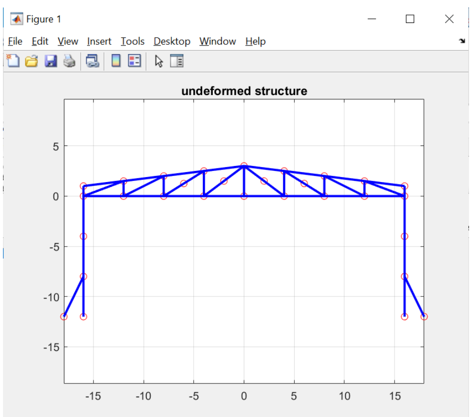
```
%after several attempts I obtained a good result by using 2 beams at both
%external sides of pillars
%considering both total mass requirement and also reducing at least 50% the
%clamping moment on the base of the left pillar
```

Before we have to modiy the input file adding news beams and new nodes:

```
*NODES
1      0 0 0      -16   0
2      0 0 0      -12   0
3      0 0 0      -8    0
4      0 0 0      -4    0
5      0 0 0       0    0
6      0 0 0       4    0
7      0 0 0       8    0
8      0 0 0       12   0
9      0 0 0      16    0
10     0 0 0      16    1
11     0 0 0       12   1.5
12     0 0 0       8    2
13     0 0 0       6 1.25
14     0 0 0       4    2.5
15     0 0 0       2 1.5
16     0 0 0       0    3
17     0 0 0     -2   1.5
18     0 0 0      -4    2.5
19     0 0 0      -6 1.25
20     0 0 0      -8    2
21     0 0 0      -12   1.5
22     0 0 0      -16   1
23     0 0 0      -16 -4
24     0 0 0       16 -4
25     0 0 0       -16 -8
26     0 0 0       16  -8
27     1 1 1       -16 -12
28     1 1 1       -17.95 -12
29     1 1 1       16 -12
30     1 1 1       17.95 -12
*ENDNODES
```

```
*BEAMS
1   1   2  30.7 8.0587e+08 8017520
2   2   3  30.7 8.0587e+08 8017520
3   3   4  30.7 8.0587e+08 8017520
4   4   5  30.7 8.0587e+08 8017520
5   5   6  30.7 8.0587e+08 8017520
6   6   7  30.7 8.0587e+08 8017520
7   7   8  30.7 8.0587e+08 8017520
8   8   9  30.7 8.0587e+08 8017520
9   9   10 30.7 8.0587e+08 8017520
10  10  11 30.7 8.0587e+08 8017520
11  9   11 30.7 8.0587e+08 8017520
12  11  12 30.7 8.0587e+08 8017520
13  8   12 30.7 8.0587e+08 8017520
14  12  14 30.7 8.0587e+08 8017520
15  7   13 30.7 8.0587e+08 8017520
16  13  14 30.7 8.0587e+08 8017520
17  14  16 30.7 8.0587e+08 8017520
18  6   15 30.7 8.0587e+08 8017520
19  15  16 30.7 8.0587e+08 8017520
20  5   16 30.7 8.0587e+08 8017520
21  16  17 30.7 8.0587e+08 8017520
22  4   17 30.7 8.0587e+08 8017520
23  16  18 30.7 8.0587e+08 8017520
24  18  19 30.7 8.0587e+08 8017520
25   3  19 30.7 8.0587e+08 8017520
26  18  20 30.7 8.0587e+08 8017520
27  2   20 30.7 8.0587e+08 8017520
28  20  21 30.7 8.0587e+08 8017520
29  21  1  30.7 8.0587e+08 8017520
30  21  22 30.7 8.0587e+08 8017520
31  1   22 30.7 8.0587e+08 8017520
32  2   21 30.7 8.0587e+08 8017520
33  3   20 30.7 8.0587e+08 8017520
34  4   18 30.7 8.0587e+08 8017520
35  6   14 30.7 8.0587e+08 8017520
36  7   12 30.7 8.0587e+08 8017520
37  8   11 30.7 8.0587e+08 8017520
38  1   23 90.7 2.3793e+09 99292000
39  23  25 90.7 2.3793e+09 99292000
40  25  28 30.7 8.0587e+08 8017520
41  25  27 90.7 2.3793e+09 99292000
42  9   24 90.7 2.3793e+09 99292000
43  24  26 90.7 2.3793e+09 99292000
44  26  29 90.7 2.3793e+09 99292000
45  26  30 30.7 8.0587e+08 8017520
*ENDBEAMS


*DAMPING
0.2 8.0e-5
```

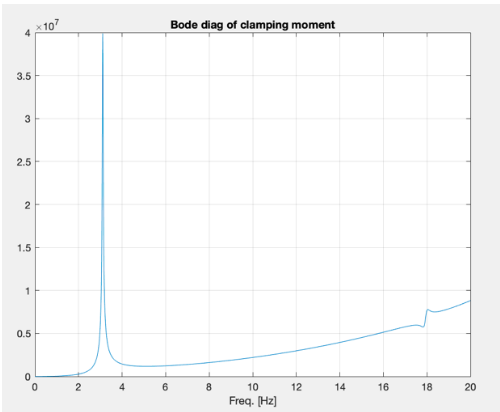
Undeformed structure obtained this way is:



Then the ***matrix partition*** has to be recomputed considering the new structure and so also the FRF of the clamping moment has to be recomputed:

```
load('punto7v4_mkr.mat')

%matrix partition M, R, K

MFF = M(1:78, 1:78);
MFC = M(1:78, 79:90);
MCF = MFC';
MCC = M(79:90, 79:90);

RFF = R(1:78, 1:78);
RFC = R(1:78, 79:90);
RCF = RFC';
RCC = R(79:90, 79:90);

KFF = K(1:78, 1:78);
KFC = K(1:78, 79:90);
KCF = KFC';
KCC = K(79:90, 79:90);

% 4. Compute and print the Bode diagrams of the frequency response functions for a unit horizontal
% displacement applied in phase at the base of both pillars, considering as output the horizontal reaction
% force and clamping moment in the clamp of the left pillar.

i=sqrt(-1);
vett_f=0:0.01:20;

f0=[1 0 0 1 0 0 1 0 0 1 0 0]';  %12 DOC

xc0=f0;
for k=1:length(vett_f)
    ome=vett_f(k)*2*pi;
    A = -ome^2*MFF+i*ome*RFF+KFF;
    Qfc = -(-ome^2*MFC + i*ome*RFC + KFC)*xc0;
    xf0=A\Qfc;  % 72 dof

    fc = (-ome^2*MCC + i*ome*RCC + KCC)*xc0 + (-ome^2*MCF + i*ome*RCF + KCF)*xf0;
    % vector size 6, so 6 constraints definition, so It is considered
    % T and M yet

    out1=fc(3);
    mod1(k)=abs(out1);
    fas1(k)=angle(out1);

end

figure
plot(vett_f,mod1);grid
title('Bode diag of clamping moment');
xlabel('Freq. [Hz]');
```

Bode diagram of the clamping moment in the new structure:



The clamping moment we obtained is reduced by a factor 5, in order to do so we increased the total mass of the 4.7% .